T3000 API & Protocol

Message: GET_PANELS_LIST · built Nov 13, 2026

Message: GET_PANELS_LIST

GET_PANELS_LIST retrieves all online panels (devices) in the network. Use this to discover available devices and their basic information.

Common Uses:

- Device discovery
- Refreshing device list
- Checking device online status
- Building device selection menus

Overview

Message Type: GET_PANELS_LIST **File Location:** `BacnetWebView.cpp` line 2503 **Purpose:** List all online panels

Request Format

JSON

```
{
  "action": "GET_PANELS_LIST",
  "source": 1
}
```

Parameters

Parameter	Type	Required	Description
`action`	String	✅	Must be "GET_PANELS_LIST"
`source`	Integer	✅	Message source: 0=T3000, 1=WebUI

Response Format

JSON

```
{
  "action": "GET_PANELS_LIST_RES",
  "data": [
    {
      "panel_number": 0,
      "object_instance": 123456,
      "serial_number": 12345,
      "online_time": 1704672000,
      "pid": 19,
      "panel_name": "Controller 1"
    },
    {
      "panel_number": 1,
      "object_instance": 123457,
      "serial_number": 12346,
      "online_time": 1704672100,
      "pid": 19,
      "panel_name": "Controller 2"
    }
  ]
}
```

Response Fields

Field	Type	Description
`panel_number`	Integer	Panel ID (0-254)
`object_instance`	Integer	BACnet object instance
`serial_number`	Integer	Device serial number
`online_time`	Integer	Last response time (Unix timestamp)
`pid`	Integer	Product ID / panel type
`panel_name`	String	Device name

Implementation

C++

```
case WEBVIEW_MESSAGE_TYPE::GET_PANELS_LIST:
{
    tempjson["action"] = "GET_PANELS_LIST_RES";

    int send_index = 0;
    for (int i = 0; i < g_bacnet_panel_info.size(); i++)
    {
        int nret = LoadOnlinePanelData(g_bacnet_panel_info.at(i).panel_number);

        if (nret > 0)
        {
            tempjson["data"][send_index]["panel_number"] =
                g_bacnet_panel_info.at(i).panel_number;
            tempjson["data"][send_index]["object_instance"] =
                g_bacnet_panel_info.at(i).object_instance;
            tempjson["data"][send_index]["serial_number"] =
                g_bacnet_panel_info.at(i).nseiral_number;
            tempjson["data"][send_index]["online_time"] =
                g_bacnet_panel_info.at(i).online_time;
            tempjson["data"][send_index]["pid"] =
                g_bacnet_panel_info.at(i).npid;
            tempjson["data"][send_index]["panel_name"] =
                (char*)g_Device_Basic_Setting[...].reg.panel_name;
            send_index++;
        }
    }

    break;
}
```

Panel Types (PID)

PID	Device Type
19	T3-BB
213	T3-8O
220	T3-LB
316	T3-TB

Online Status

online_time values:

- `0` - Never connected (offline)
- `> 0` - Unix timestamp of last response

Determining if online:

```javascript
const isOnline = (panel) => {
  const now = Math.floor(Date.now() / 1000);
  const diff = now - panel.online_time;
  return diff < 300; // Online if responded within 5 minutes
};
```

Performance

Scenario	Time
1-5 panels	< 100ms
10-20 panels	100-300ms
50+ panels	300-500ms

Usage Example

TYPESCRIPT

```typescript
// Request panels list
const response = await fetch('/api/webview/message', {
  method: 'POST',
  body: JSON.stringify({
    action: 'GET_PANELS_LIST',
    source: 1
  })
});

const data = await response.json();

// Filter online panels
const onlinePanels = data.data.filter(panel => {
  const now = Math.floor(Date.now() / 1000);
  return (now - panel.online_time) < 300;
});

// Build device list UI
onlinePanels.forEach(panel => {
  console.log(`${panel.panel_name} (SN: ${panel.serial_number})`);
});
```

See Also

- GET_PANEL_DATA - Load specific panel data
- GET_ENTRIES - Query available entries
- Control Messages Index